

2024



Reimagining the Patient Experience using AWS with the Froedtert & the MCW Health Network

Learn how the Froedtert & the Medical College of Wisconsin health network is transforming the patient experience using AWS.

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70,000

patients served monthly through digital services

Developed

personalized and patient-centric applications

Deployed

digital applications at scale quickly and



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Overview

The [Froedtert & the Medical College of Wisconsin](#) (MCW) health network is using technology to reimagine healthcare. Its innovation arm, [Inception Health](#), seeks to improve the patient experience through digital applications that use artificial intelligence (AI) and data analytics to personalize care and alleviate stress on clinicians. "In the same way that we build and own our physical care experiences such as hospitals, urgent care centers, and emergency rooms, we need to own our technology," says Melek Somai, MD, MPH, vice president and chief technology and chief product officer at Inception Health and Froedtert & MCW. "We want technology to become part of our DNA."

In its quest to meaningfully transform digital engagement, Inception Health built a digital platform powered by Amazon Web Services (AWS). Patients can now access, control, and understand their own data, and state-of-the-art applications provide insights and corresponding personalized programs that facilitate access to care.



Opportunity | Using AWS to Transform the Patient Experience for the Froedtert & MCW Health Network

With more than 1.2 million yearly visits to its network of more than 2,300 physicians, the Froedtert & MCW health network supports a mission of patient care, innovation, medical research, and education. Healthcare systems typically use technology and analytics for supporting operations, such as population health programs and revenue-cycle or supply-chain management. But the Froedtert & MCW health network wanted to transform the frontend patient experience of healthcare, so it created Inception Health in 2015. A team of researchers, data scientists, software engineers, and product designers led the digital transformation incorporating AWS to build a nimble structure to connect all the different components of an increasingly disparate digital journey. "We want to improve the patient experience for patients and to forge our path in the digital health space," says Dr. Bradley Crotty, chief digital officer



for the Froedtert & MCW health network and interim president of Inception Health. “We made a very intentional strategic decision to use AWS because of its focus on consumer-oriented technologies.”

As the COVID-19 pandemic strained healthcare resources, the Froedtert & MCW health network was able to quickly deploy telehealth and asynchronous care at scale. “We were not in a position to build our digital engagement platform from the ground up in our data center,” says Crotty. “We needed to be a very lean team, relying on automation and leaning heavily on cloud-centered tools and technologies.” Working alongside [AWS Professional Services](#), a global team of experts that helps organizations achieve their desired business outcomes using AWS, the Inception Health team built a resilient infrastructure that functions with 99.999 percent availability.

“We knew that we could build a HIPAA-compliant infrastructure in the cloud effortlessly using AWS services,” says Somai. “Working alongside AWS Professional Services and the AWS healthcare team, we stood up an entire multi-account organization in a couple of weeks.”

Solution | Serving 70,000 Patients Monthly through a Modern, Secure Infrastructure

Inception Health’s small team quickly deploys highly reliable, secure digital products that personalize care for patients, using the [Shared Responsibility Model](#) for the security and compliance of its infrastructure. The serverless architecture uses [AWS Lambda](#), a compute service that runs code in response to events and automatically manages the compute resources. An API layer helps to add patient-centric applications at scale. “Our developers build the entire user journey—infrastructure and logic—within the same code,” says Somai. “That removes some great complexities in healthcare: orchestration, interoperability, and integration.”

The team uses tools for multimodal data analysis through [Amazon SageMaker](#), which lets organizations build, train, and deploy machine learning models for virtually any use case with fully managed infrastructure, tools, and workflows. For example, Inception Health uses analytics to identify patients who need preventive care, then sends personalized recommendations through [Amazon Pinpoint](#)—which offers organizations one customizable tool to deliver customer communications across channels, segments, and campaigns at scale.

To collect its data to power machine learning and other applications, Inception Health uses [Amazon Firehose](#), an extract, transform, load service that reliably captures, transforms, and delivers streaming data into data lakes, warehouses, and analytics services. “We conduct a lot of our discovery by listening to our customers,” says Somai.



Serving 70,000 patients monthly, the Froedtert & MCW health network is moving toward a self-service framework with guardrails to keep patients safe. For example, it offers a depression screening that patients access privately through the mobile app. “We’re using technology not only to replicate today’s experience but also to reimagine the services and the value that a healthcare system could deliver to its community through technology,” Crotty says. “Our technology helps us absorb information and build experiences to meet our patients’ needs.”

Outcome | Continuing to Transform Healthcare through AI in the Cloud

Inception Health plans to become an early adopter of generative AI to further enhance the patient experience. Specifically, it is looking into [Amazon Bedrock](#)—a fully managed service that offers a choice of high-performing foundation models through a single API, along with a broad set of capabilities that organizations need to build generative AI applications with security, privacy, and responsible AI.

“Quality care today means availability and access, and that’s the power for us to transcend and transform the healthcare system,” says Somai. “Using AWS, we want to empower patients to make the right decisions and remove the healthcare system as a gatekeeper.”

About Froedtert & MCW

With 10 hospitals, more than 2,300 physicians, and more than 45 health centers, the Froedtert & the Medical College of Wisconsin (MCW) health network is a partnership between Froedtert Health and MCW.

AWS Services Used

Amazon SageMaker

Amazon SageMaker is a fully managed service that brings together a broad set of tools to enable high-performance, low-cost machine learning (ML) for any use case.

[Learn more »](#)

AWS Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

[Learn more »](#)



Amazon Kinesis Data Firehose

Amazon Data Firehose provides the easiest way to acquire, transform, and deliver data streams within seconds to data lakes, data warehouses, and analytics services.

[Learn more »](#)

Amazon Bedrock

Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies via a single API, along with a broad set of capabilities you need to build generative AI applications with security, privacy, and responsible AI.

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